

GradeSense Assessment - Answer Sheet

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Total marks: 15

Answer 1. (Science)

A simple electric circuit is a closed conducting path through which current can flow. The battery provides the potential difference that drives the current, and the switch opens or closes the circuit. When the switch is closed the path is complete and current flows through the bulb and the resistor.

The battery, switch, resistor, bulb and ammeter are connected in series in the main circuit. The ammeter is connected in series because it measures the current flowing through the circuit. The voltmeter is connected in parallel across the bulb because it measures the potential difference between the two ends of the bulb.

The conventional current flows from the positive terminal of the battery through the external circuit and back to the negative terminal, and this direction is marked on the diagram.

By Ohm's law, $V = IR$. If the voltage of the battery stays constant and the resistance is increased, the current flowing through the circuit decreases. Reducing the resistance allows more current to flow.

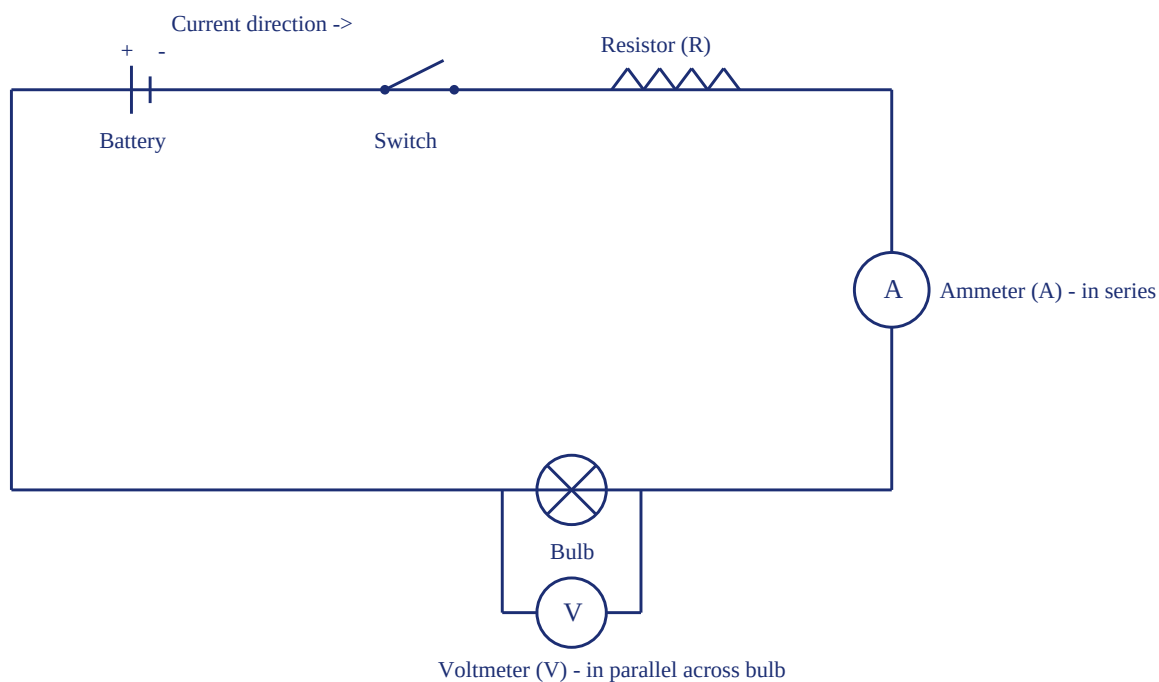


Fig. 1 - Circuit diagram

Answer 2. (English)

Technology has made information far easier to reach, but I believe it only makes students better learners when it is used to support their own thinking rather than replace it.

Access to digital libraries and video explanations genuinely helps. A student who cannot follow one explanation of refraction can watch three more until one makes sense, which a single classroom lesson cannot offer. However, the same access allows a student to search for a solution before attempting the problem, and a student who never struggles with a question never develops the reasoning to answer the next one.

It is fair to argue that easy access simply saves time, and that the time saved can be spent on deeper study. That is true for a student who already knows how to study, but for most students the saved time is not reinvested; the answer is copied and the reasoning is skipped.

Technology is therefore best understood as a tool that supports thinking. Used to check and extend your own reasoning it makes learning far better; used as a substitute for reasoning it makes learning worse.

Answer 3. (Economics)

The graph is drawn with quantity on the horizontal axis and price on the vertical axis. The demand curve slopes downward from left to right and the supply curve slopes upward from left to right.

The two curves intersect at a price of Rs. 30 and a quantity of 60 units. This is the market equilibrium because at that price the quantity demanded is equal to the quantity supplied.

If the price is below the equilibrium price, quantity demanded exceeds quantity supplied and there is a shortage, which pushes the price up. If the price is above the equilibrium price, quantity supplied exceeds quantity demanded and there is a surplus, which pushes the price down.

If the cost of production increases, producers supply less at every price, so the supply curve shifts to the left. The new equilibrium is at a higher price and a lower quantity, assuming demand is unchanged.

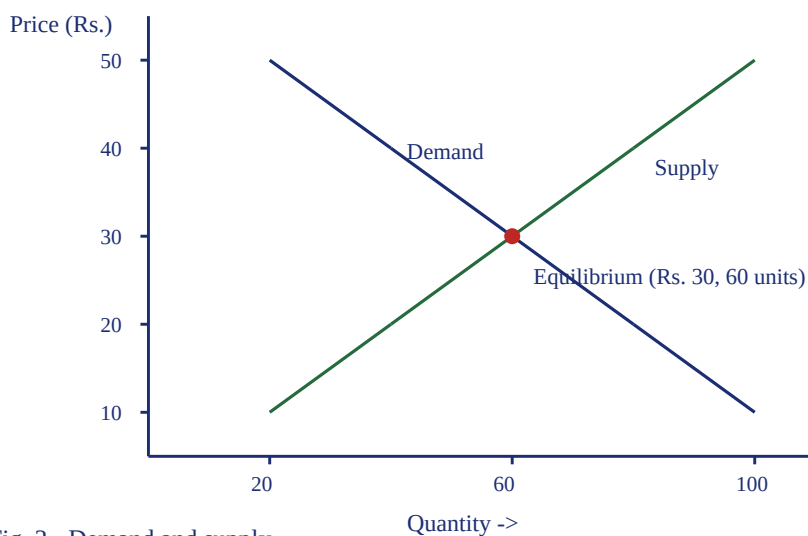


Fig. 2 - Demand and supply